1. $x - 2y + 3z = 3$ and $2x + y + 5z = 8$ and $3x - y - 3z = -22$

   $x - 2y + 3z = 3$

   $+ 4x + 2y + 10z = 16$

   $5x + 13z = 19$

   $2x + y + 5z = 8$

   $3x - y - 3z = -22$

   $5x + 2z = -14$

   $5x + 13z = 19$

   $5x + 2z = -14$

   $11z = 33$

   $z = 3$

   $x - 2y + 3(3) = 3$

   $4x + 2y + 10(3) = 16$

   $5x + 39 = 19$

   $5x = -20$

   $x = -4$

   $2(-4) + y + 5(3) = 8$

   $-8 + y + 15 = 8$

   $y + 7 = 8$

   $y = 1$

2. $3x + 2y - z = 10$ and $x + 4y + 2z = 3$ and $2x + 3y - 5z = 23$

   $4x + 6y + z = 13$

   $4x + 6y - 10z = 46$

   $11z = -33$

   $z = -3$

   $3x + 2y = 7$

   $3x + 12y = 27$

   $10y = 20$

   $y = 2$

   $3x + 4 + 3 = 10$

   $3x + 7 = 10$

$3x = 3$

$x = 1$

3. $5x - 4y + 3z = 15$ and $6x + 2y + 9z = 13$ and $7x + 6y - 6z = 6$

   $15x - 12y + 9z = 45$

   $6x + 2y + 9z = 13$

   $9x - 14y = 32$

   $15z - 12y + 9z = 45$

   $105x + 9y - 9z = 9$

   $255x - 3y = 54$

   $y = 85x - 18$

   $9x - 14(85x - 18) = 32$

   $9x - 119x - 252 = 38$

   $-110x - 252 = 38$

   $-110x = 290$

   $x = 2$

   $y = 85(2) - 18$

   $y = 17 - 18$

   $y = -1$

   $5(2) - 4(-1) + 3z = 15$

   $10 + 4 + 3z = 15$

   $14 + 3z = 15$

   $3z = 1$

   $z = 13$

4. $5x - 4y - 6z = 21$ and $-2x + 3y + 4z = -15$ and $3x - 7y - 5z = 15$

   $-6x + 9y + 12z = -45$

   $6x - 14y - 10z = 30$

   $-5y + 2z = -15$

   $10x - 8y - 12z = 42$

   $-10x + 15y + 20z = -75$

   $7y + 8z = -33$

   $20y - 8z = 60$

   $27y = 27$

   $y = 1$

   $-5(1) + 2z = -15$

   $2z - 5 = -15$

$2z = -10$

$z = -5$

$5x - 4(1) - 6(-5) = 21$

$5x - 4 + 30 = 21$

$5x + 26 = 21$

$5x = -5$

$x = -1$

5. The one with missing variables, I'm too tired to right the equations

$2y = 3z - 8$

$y = 4 - 15z$

$3x + 4(4 - 15z) = 19$

$3x + 16 - 6z = 19$

$3x - 6z = 3$

$x - 2z = 1$

$x = 1 + 2z$

$4(1 + 2z) - 5z = 7$

$4 + 8z - 5z = 7$

$4 + 3z = 7$

$3z = 3$

$z = 1$

$x = 1 + 2(1)$

$x = 3$

$2y + 3z = 8$

$2y + 3(1) = 8$

$2y + 3 = 8$

$2y = 5$

$y = 25$

6. It is inconsistent because the first and second equations contradict eachother